



SOC DETECTION LAB

From Attack to Alert

Simulating a Live Intrusion and Detecting It in Real Time with Splunk

Kali Linux • Nmap • Hydra • Splunk SPL

The Role of a SIEM — Why Splunk

A Security Information and Event Management (SIEM) platform delivers centralized security monitoring across the entire IT infrastructure. Splunk performs this role through three key stages:



01

Log Collection

Gathers logs simultaneously from multiple devices, endpoints, servers, and network equipment across the environment.



02

Log Aggregation

Centralizes massive volumes of logs into a single repository, ordering them chronologically instead of leaving them scattered.



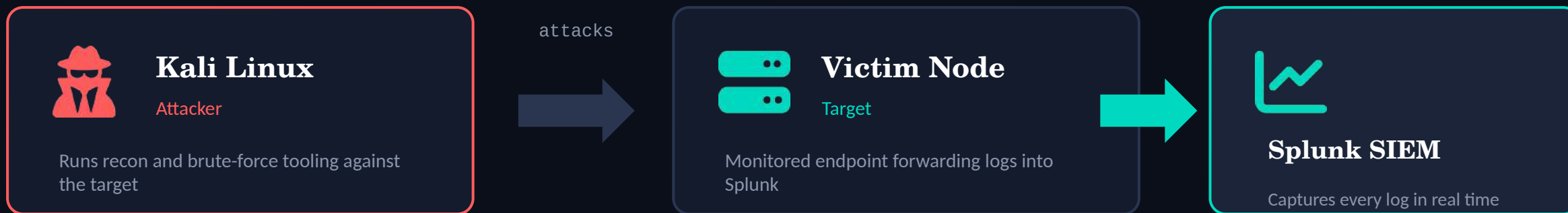
03

Normalization

Parses raw logs from different vendors and OSs (e.g. Windows vs. Linux) into one unified, structured format for seamless search.

Attack Scenario — Attacker Side

The simulation runs from a Kali Linux machine acting as the attacker, targeting a victim node on the same network.



**PHASE 1**
Reconnaissance
Host & port discovery using Ping and Nmap.

**PHASE 2**
Exploitation
Brute-force credential attack against SSH using Hydra.

Reconnaissance — Host & Port Discovery



Ping Sweep

A ping is sent to verify that the victim node is alive and reachable on the network.



Nmap Port Scan

Nmap (Network Mapper) scans the target to identify open ports, running services, and the operating system.



Splunk captured packet-flow anomalies and port-scanning signatures — a high frequency of connection requests across many ports within a short timeframe.



kali@attacker: ~

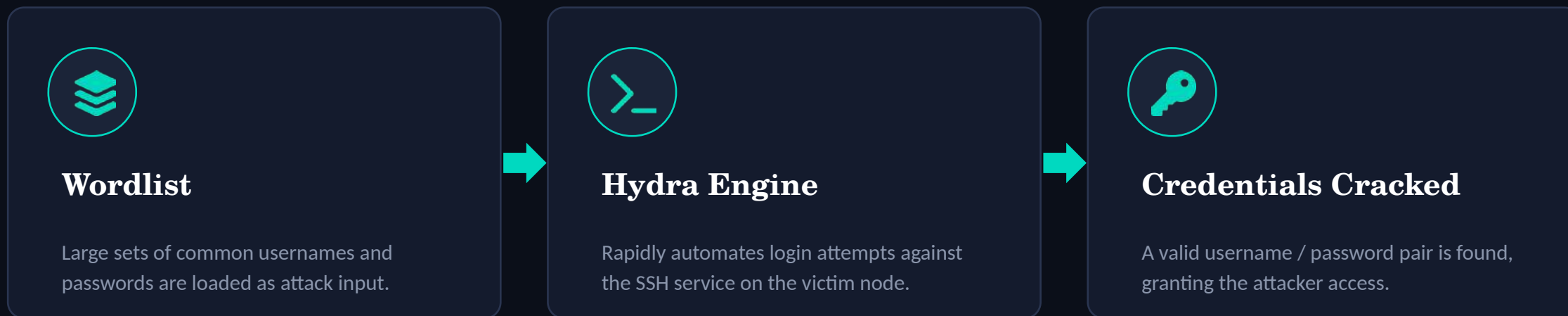
```
$ ping 10.10.10.15  
64 bytes from 10.10.10.15: icmp_seq=1 ttl=64
```

```
$ nmap -sV -O 10.10.10.15  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.4  
80/tcp    open  http     Apache 2.4.41  
3306/tcp  open  mysql    MySQL 8.0
```

```
OS: Linux 5.x | 18 ports scanned
```

Exploitation — Brute-Force Attack

Targeting the SSH service identified during reconnaissance, the attacker uses Hydra to conduct a brute-force credential attack — automatically and rapidly testing a wordlist of usernames and passwords until access is gained.



Detected: Splunk logged a sudden spike of Failed Login Attempts from the attacker's IP within seconds, followed by a Successful Login event — the definitive indicator of compromise (IoC).

Event Detection & Telemetry

Every phase executed on Kali Linux was captured and recorded in real time within Splunk.



Detecting Reconnaissance

Ping & Nmap

Splunk captured packet-flow anomalies and port-scanning signatures — a high frequency of connection requests to many different ports within a short timeframe.

SIGNAL

High-frequency multi-port connection attempts



Detecting Brute Force

Hydra

Splunk logged a sudden spike of Failed Login Attempts originating from the attacker's IP address within seconds.



Successful Login logged immediately after — the definitive Indicator of Compromise (IoC).

Detecting Brute-Force Attacks

Failed logins spike from a single source

>_ search.spl

```
sourcetype=linux_secure "Failed password"  
| stats count by src_ip, user  
| where count > 10
```



How it works

Filters authentication logs for failed-password events.

Groups attempts by source IP and username.

Flags any pairing that crosses a 10-attempt threshold — a hallmark of automated tools like Hydra.

Detecting Port Scanning / Recon

Unusually broad port coverage from one host

>_ search.spl

```
sourcetype=ids OR firewall  
| stats dc(dest_port) as unique_ports by src_ip  
| where unique_ports > 15
```



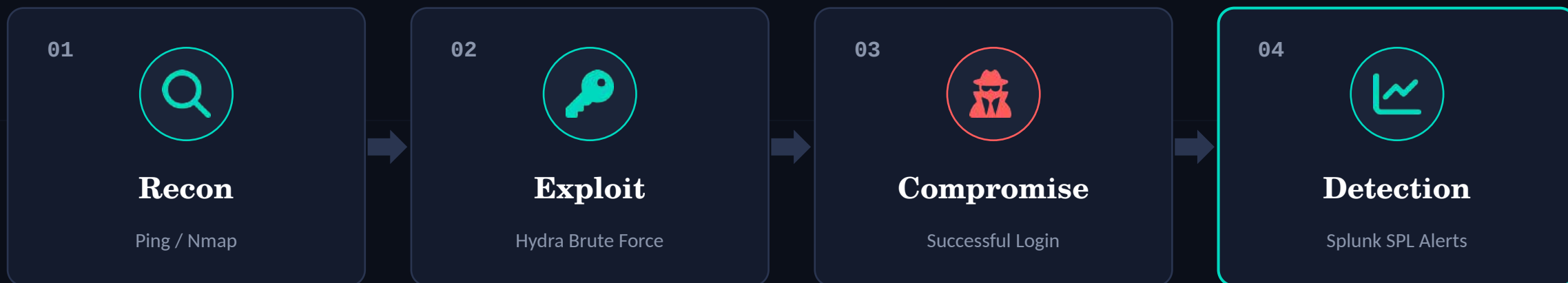
How it works

Pulls events from IDS and firewall sources.

Counts the distinct destination ports each source IP touched.

Flags hosts contacting more than 15 unique ports — consistent with an Nmap-style scan.

The Full Attack Chain — Seen End to End

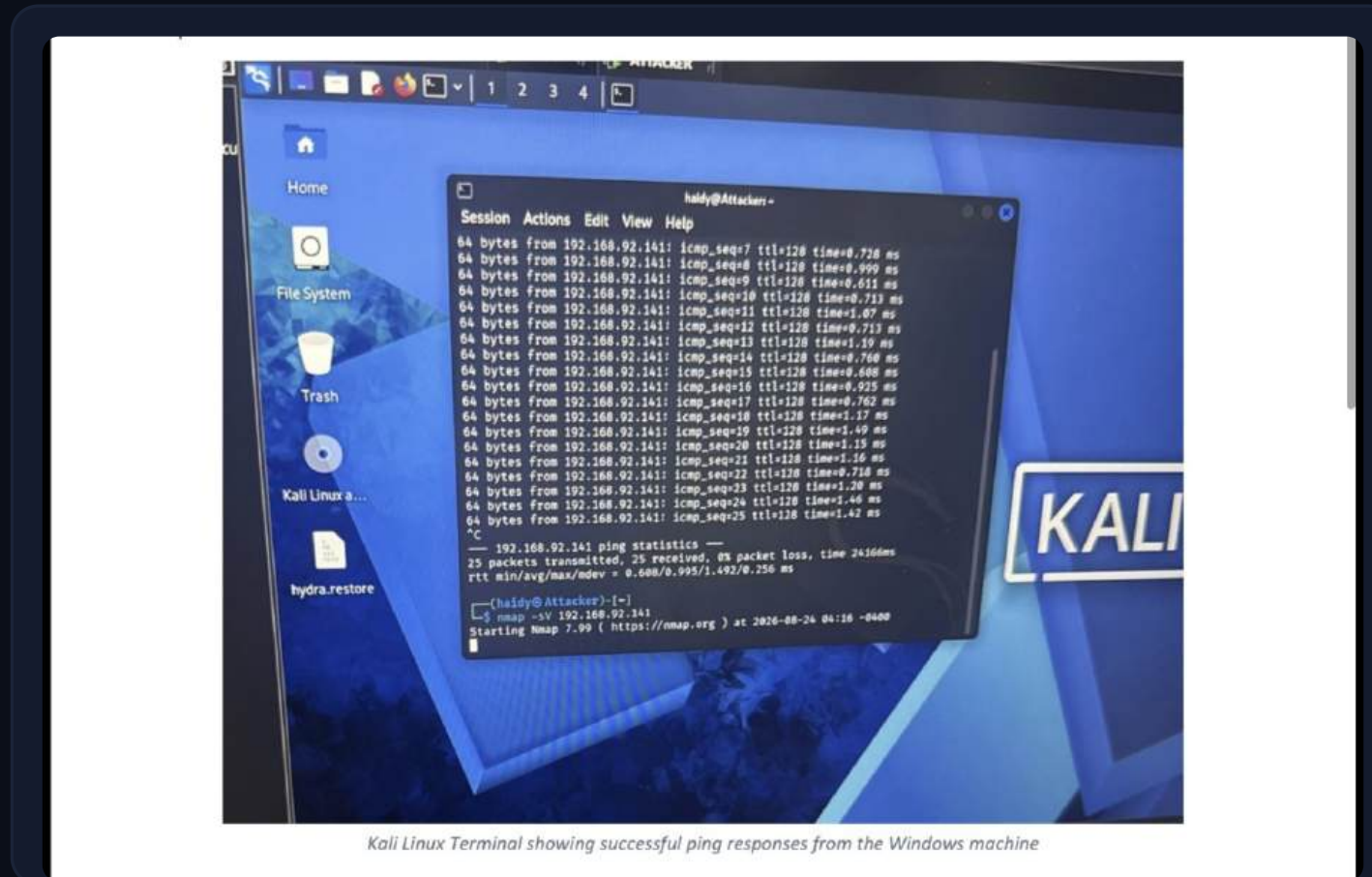


Key Takeaway

Normalized, centralized telemetry let a SOC analyst trace an attack from the first port probe to the final compromised login — turning scattered raw logs into a clear, actionable timeline.

Reconnaissance — Real Lab Screenshot

Captured from the actual Kali Linux attacker VM during the live exercise

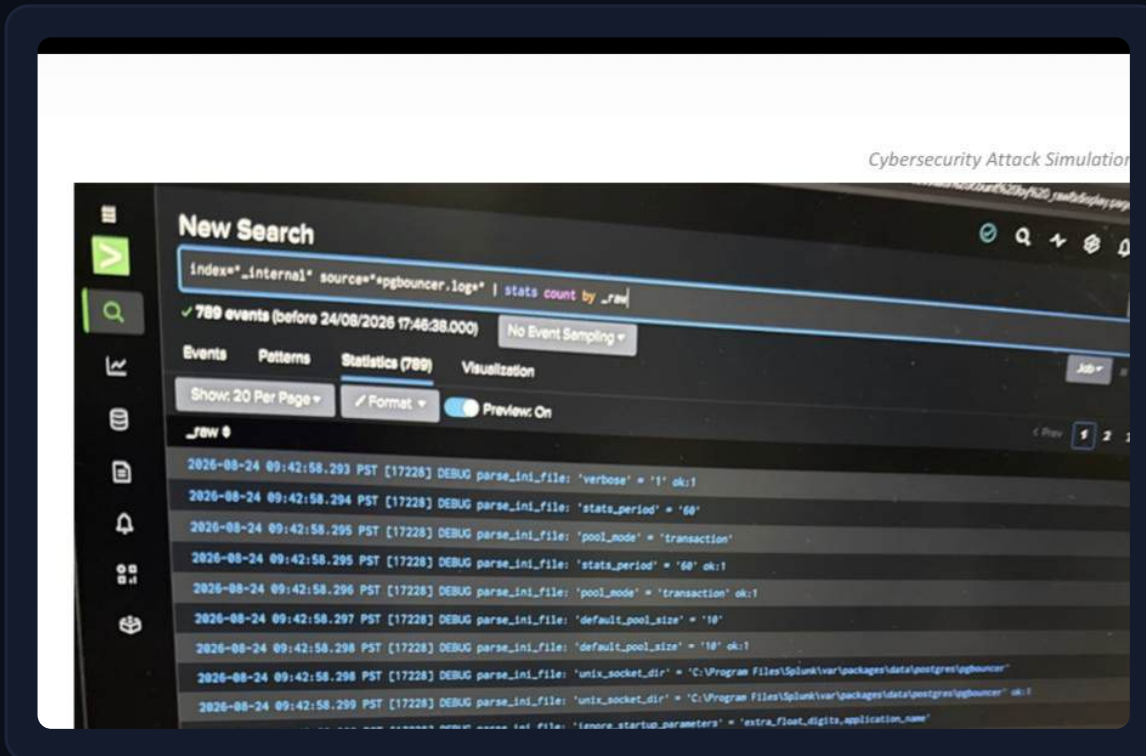


Kali Linux Terminal showing successful ping responses from the Windows machine

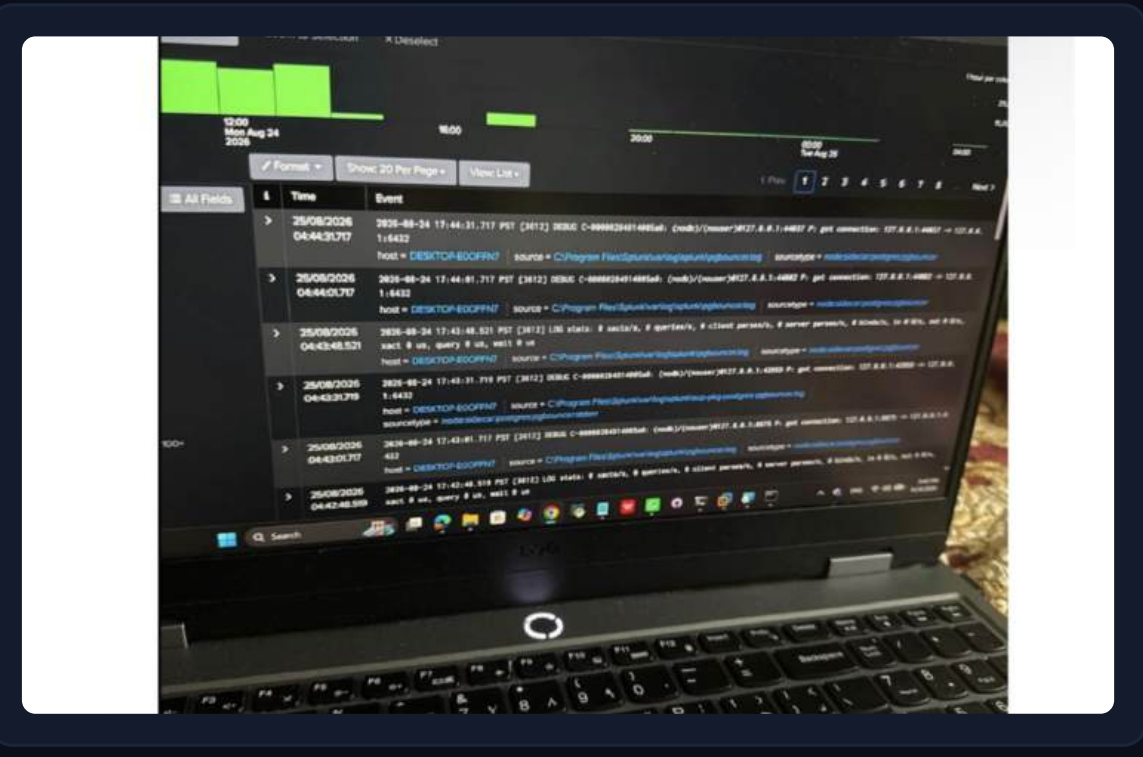
Kali Linux terminal: successful ping sweep to 192.168.92.141, followed by the nmap -sV scan launch

Detection — Real Splunk Screenshots

Captured from the actual Splunk instance monitoring the victim node during the attack



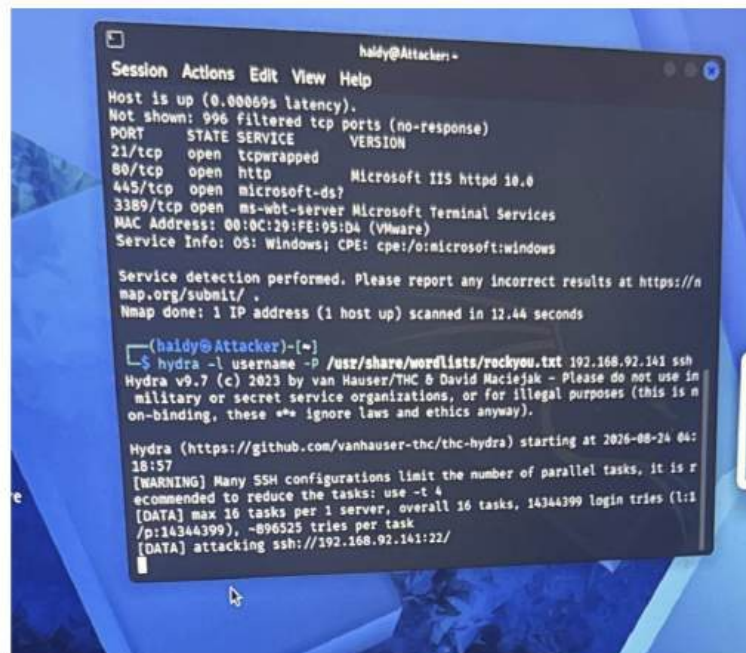
Splunk Statistics view — pgbouncer log events aggregated with 'stats count by _raw'



Splunk event timeline showing spikes in event volume during the attack window

Exploitation — Real Hydra Attack

Captured from the actual Kali Linux attacker VM during the brute-force phase



```
haidy@Attacker:~$ nmap -sC -sV -p- 192.168.92.141
Nmap scan results showing open ports and service versions, followed by the Hydra command starting the brute-force attempt

Session Actions Edit View Help
Host is up (0.00000s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  tcpwrapped
80/tcp    open  http      Microsoft IIS httpd 10.0
445/tcp   open  microsoft-ds?
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
MAC Address: 00:0C:29:FE:95:D4 (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

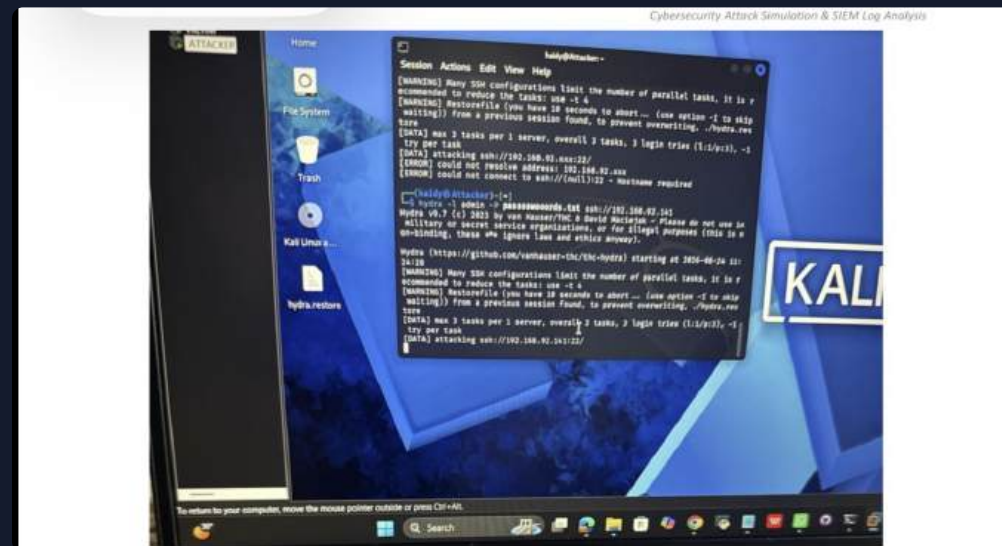
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 12.44 seconds

(haidy@Attacker)~$ hydra -l username -P /usr/share/wordlists/rockyou.txt 192.168.92.141 ssh
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is on-binding, these *do* ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-24 04:18:57
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (1:1 /p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.92.141:22/
```

Nmap scan results showing open ports and service versions, followed by the Hydra command starting the brute-force attempt

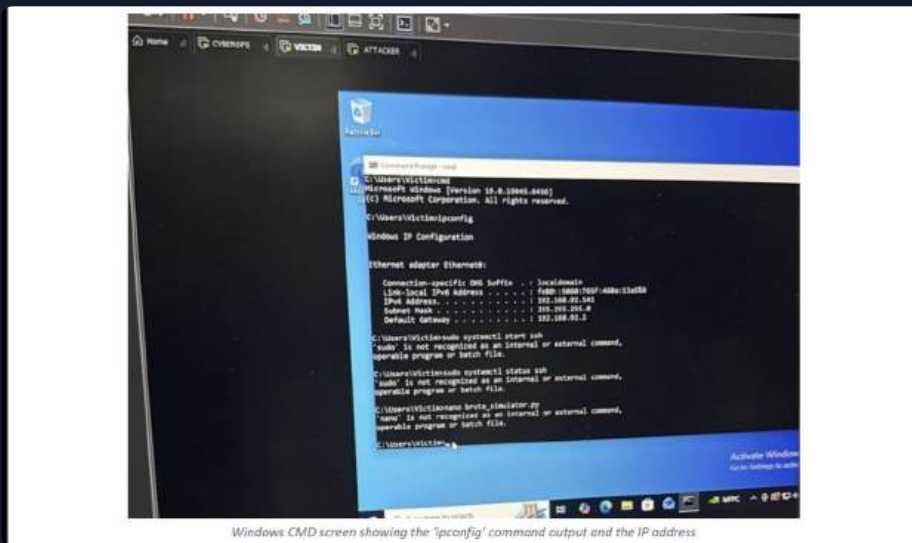
Nmap scan results (open ports 21/80/445/3389) followed by the Hydra command launch



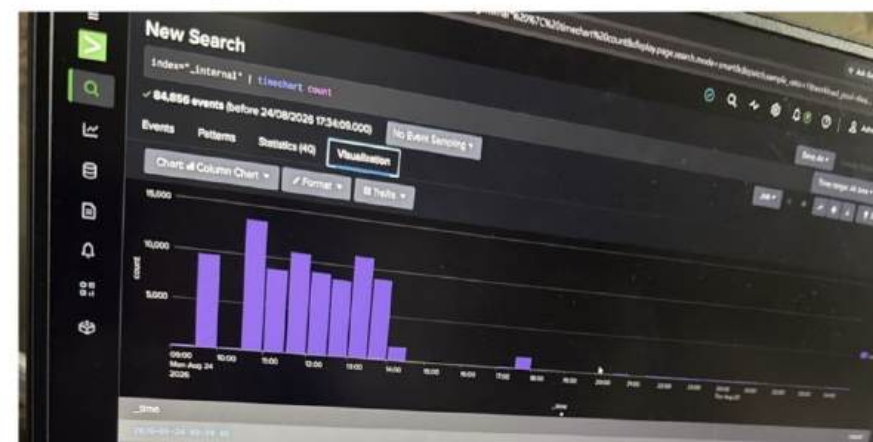
Hydra actively brute-forcing SSH credentials against the victim node

Target & Detection — Additional Evidence

Victim machine network configuration and the full Splunk event-volume timechart



Windows CMD 'ipconfig' output confirming the victim's IP address (192.168.92.141)



Splunk timechart — 84,856 events across index="_internal", showing the attack spike